# Fatemeh Maddahzadeh

AI Engineer - LLM, RAG & Agentic Systems
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## Profile

AI Engineer with nearly 3 years on a production, multi-tenant LLM platform where I was the founding backend engineer: RAG on PostgreSQL/pgvector, multi-provider LLM integration, tool calling and MCP, multi-step agent pipelines, streaming and LLM guardrails. Built on 6+ years of backend engineering in Node.js/TypeScript, microservices and cloud (GCP, Kubernetes), including experience as a Backend Team Leader. MSc thesis at Politecnico di Torino on routing and drift detection in multi-LLM systems (Python, LiteLLM). Available immediately.

## Professional Experience

**- Software Engineer - AI/LLM Platform at Glue Labs**

*Mar. 2023 - Dec. 2025 | Turin, Italy | Part-time, alongside MSc | Pupau (pupau.ai): a multi-tenant B2B platform where companies build AI assistants and agents on their own knowledge bases, with a public assistant marketplace.*

- **Ownership:** founding backend engineer; built the core RAG platform alone for its first ~6 months, then owned the AI core (query pipeline, LLM and embedding adapters, agents, guardrails, MCP) as the team grew to up to 7 developers; ~1,650 commits, 2nd-highest contributor
- **RAG & retrieval:** document and web ingestion with async embedding jobs on Google Cloud Tasks; PostgreSQL + pgvector (1536-dim embeddings, IVFFlat index, inner-product similarity search with per-assistant thresholds); tiktoken chunking with versioned chunks (order, page, active/outdated) for zero-downtime re-indexing
- **Embeddings:** pluggable embedding layer supporting OpenAI text-embedding-3-small/-large, ada-002 and open-source BAAI/bge-m3, served via OpenAI and Azure OpenAI deployments, with a provider factory for default selection and load balancing
- **Multi-provider LLM layer:** designed one interface over OpenAI, Azure OpenAI, Anthropic, Google Gemini, Vertex AI, Groq and Mistral, with per-call cost tracking and Prometheus metrics on provider rate-limit headroom
- **Tool calling & MCP:** built tool calling across 5 tool types (remote HTTP, internal pipelines, native tools, MCP servers/tools) and created the MCP integration (client + per-assistant configuration); RxJS execution with retries, structured error results returned to the model and human-approval steps
- **AI agents:** created the actor-based agent framework, later grown by the team to ~30 actors (LLM, knowledge base, web search, E2B code interpreter, Playwright browser use, database, router, conditional blocks) chained into multi-step pipelines; added agent-to-agent tagging
- **Streaming & quality:** Server-Sent Events streaming with persistence and optional encryption; LLM guardrails on input and output; a closed reply mode that refuses to answer without relevant retrieval (fewer hallucinations); user feedback and LLM call logs
- **Backend & infra:** NestJS, TypeORM and RxJS services on GCP (GKE, App Engine); set up the first staging and production environments; versioned DB migrations, multi-tenant API-key access

## AI Research - MSc Thesis

**- Handoff-Aware Routing and Drift Detection for Multi-LLM Systems**

*2026 | Politecnico di Torino | Python | LiteLLM | LLM evaluation*

- Built a Python middleware on LiteLLM that detects mid-conversation model switches (handoffs) and applies mitigation strategies to limit performance drift
- Designed an evaluation of 4 mitigation strategies against an unmitigated baseline across 400 trials with 5 LLMs from 5 vendors, using code-based validators and an LLM-as-a-judge
- **Best strategy (Validation-and-Retry):** mean validator score 0.787 vs 0.735 baseline, full-pass rate 74% vs 67%, Factual Stability passes 12/20 to 19/20
- Showed that no strategy wins on every axis and that code validators revealed differences the LLM judge missed, so strategies should be chosen by the quality axis a deployment most needs to protect

## Earlier Experience

**- Backend Team Leader at Lamasoo**

*Jan. 2022 - Sep. 2022 | Tehran, Iran | Online channel manager for the hospitality industry, serving hotels and agencies.*

- **Led a team of 3 backend developers:** hiring, mentoring, task assignment and progress reviews
- Planned sprints in Scrum, ran stand-ups and reviews, and conducted code reviews
- Designed scalable microservices and worked with the product team on priorities and fast bug resolution

**- Software Engineer at Lamasoo**

*Jun. 2020 - Jan. 2022 | Tehran, Iran*

- Designed and implemented microservices in Node.js/TypeScript and refactored the codebase towards a microservices architecture
- Developed RESTful APIs with front-end developers and optimized database queries and performance

**- Software Engineer at Linkap**

*Feb. 2019 - Jun. 2020 | Tehran, Iran | IoT start-up in the smart home industry.*

- Developed the Node.js backend of a smart home app, integrating IoT devices (LoRaWAN, Zigbee) in real time with MongoDB

## Education

**- Politecnico di Torino**

*Sep. 2022 - Nov. 2026 (expected) | Turin, Italy | MSc in Software Applications*

- GPA 25/30 | Thesis on multi-LLM systems (see AI Research)

**- University of Tehran**

*Sep. 2016 - Feb. 2021 | Tehran, Iran | BSc in Computer Engineering (Software Engineering)*

- **GPA 16.5/20 (last two years:** 17.36/20); ranked the top university in Iran by U.S. News
- Ranked 61st in the national university entrance exam

## Selected Project

**- Mismatch Desk | github.com/fatemeh76m/mismatch-desk | AI-assisted invoice discrepancy review | Portfolio project**

- Implemented OpenRouter structured extraction in a FastAPI service with exact source quotes; missing or ambiguous fields are flagged for human review.
- Built a LangGraph workflow with deterministic checks against purchase orders and delivery records; persists cases and append-only human decisions in PostgreSQL with Alembic.
- Added text-based PDF extraction with pypdf and a Next.js review UI. Uses synthetic data; no automatic approval or payment.

## Technical Skills

- **AI / LLM:** RAG, embeddings, vector search (pgvector), tool/function calling, MCP, AI agents and multi-step pipelines, prompt engineering, LLM guardrails, LLM evaluation, LiteLLM, tiktoken, SSE streaming, LangGraph, OpenRouter, evidence-grounded structured extraction, human-in-the-loop review
- **LLM providers & models:** OpenAI, Azure OpenAI, Anthropic, Google Gemini, Vertex AI, Groq, Mistral, BAAI/bge-m3
- **Languages:** TypeScript, JavaScript (Node.js), Python
- **Backend:** FastAPI, pypdf, NestJS, Express.js, TypeORM, RxJS, REST APIs, microservices, background jobs and message queues
- **Frontend:** HTML, CSS, React, Next.js
- **Data:** PostgreSQL (pgvector), MongoDB, Redis, SQL, Alembic migrations
- **Cloud & DevOps:** Google Cloud Platform (GKE, App Engine, Cloud Build, Cloud Tasks), Kubernetes, Docker, Prometheus, Git, Linux
- **Other:** RabbitMQ, MQTT, Scrum, Jira
- **AI-assisted development:** Codex, Cursor

## Languages

Persian: native | English: Advanced | Italian: Intermediate